



Multimodal Fusion Strategies in Remote Sensing:

Deep Learning for Detecting Marine Pollutants
and Sea Surface Features

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Project goals

Identification of marine pollution through marine debris and oil spills.

- Considering multiple features (15)
- Pre-Processing
- Adopting a semantic segmentation framework



Marine Debrise

It uses the MARIDA dataset, which distinguishes litter from various sea features, and provides additional information.

Oil Spill

Detecting one thematic class of oil spills rather than differentiating between various types of oil.

Overview

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Dataset

Dataset description

02

Pre-Processing

Refining and Pre-Processing

03

Models

MariNeXt & Ticino

04

Fusion Strategies

Early, Middle & Late Fusion

05

Results

Compare Results and Predictions

06

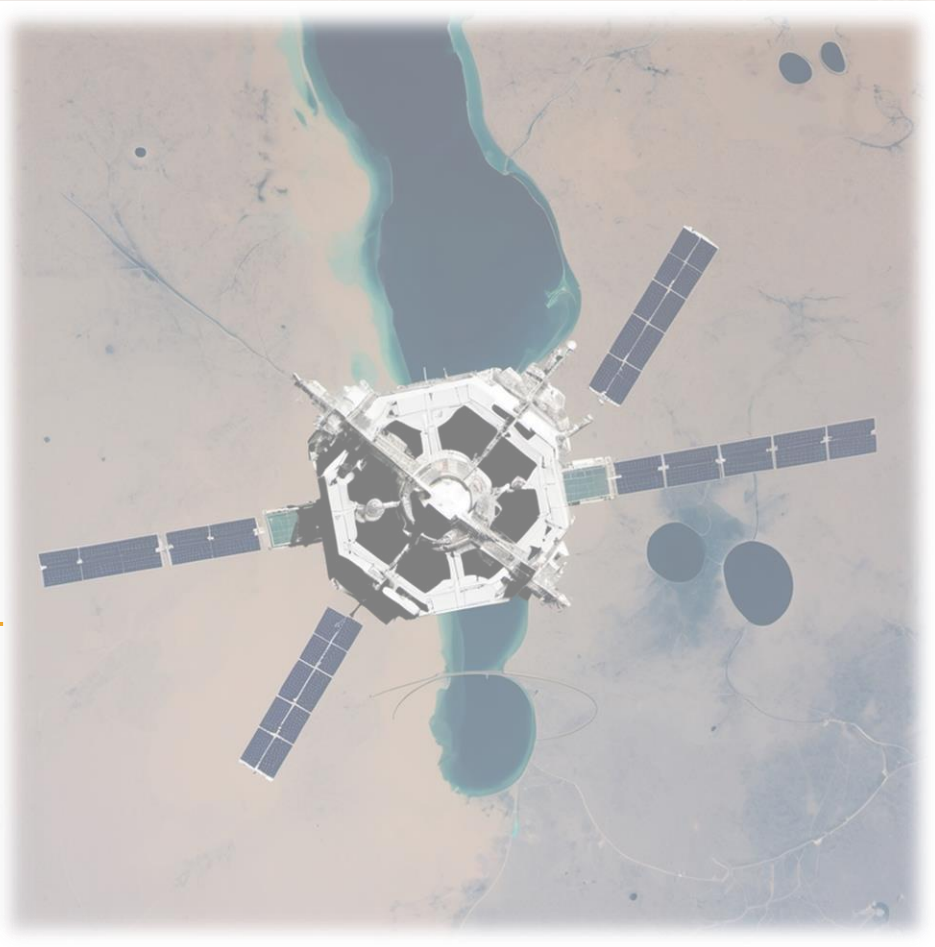
Conclusions

Final reviews and Future work

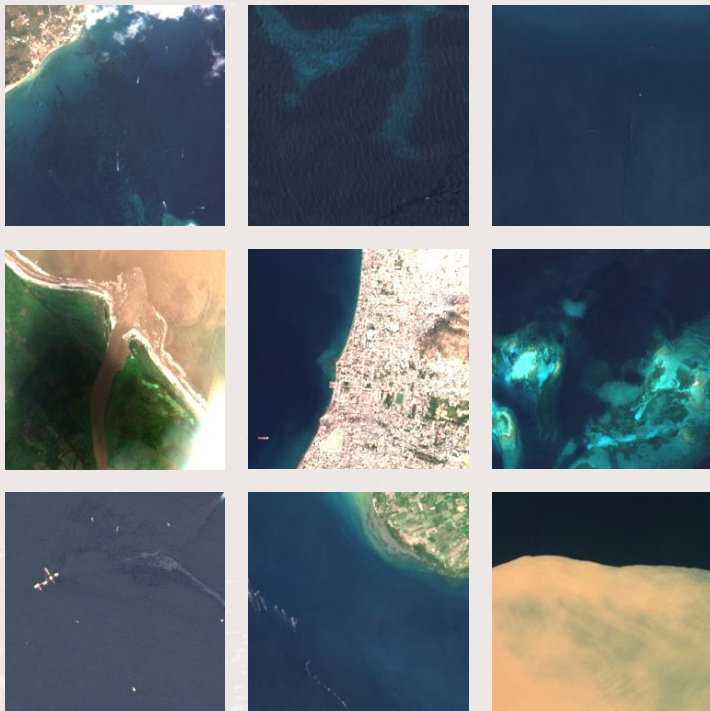
01

Dataset

Dataset description



Dataset description



The data were obtained from the **Sentinel-2** satellite:

- ❖ **174 scenes** divided in **2803 patch**.
- ❖ Collected **from 2015 to 2022**.
- ❖ **22 countries**
- ❖ **15 different classes**

❖ Three types of quality for each band:

High-quality

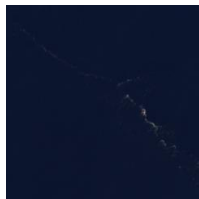
Medium-quality

Low-quality

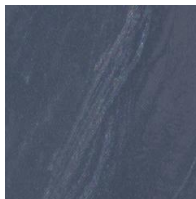
Classes

Show all types
of classes used.

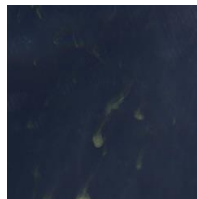
Marine Debrise



Oil Spill



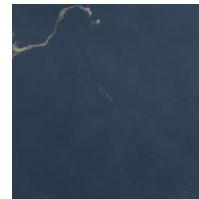
Algae



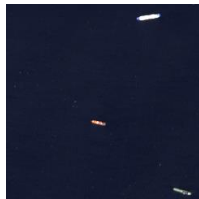
Land



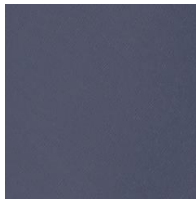
Natural Organic
Material



Ship



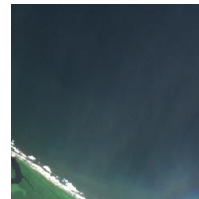
Marine Water



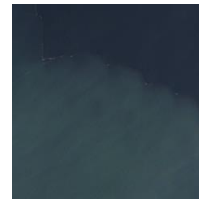
Sediment-Laden
Water



Foam



Turbid Water



Shallow Water



Wakes



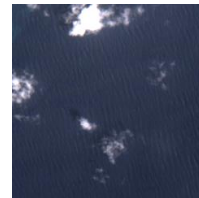
Oil Platform



Jellyfish



Clouds



02

Refining and Pre-Processing

Annotations, Mask &
Class imbalance



Refining and Pre-Processing



Annotations

Evaluation of old GTs, merging of classes, identification of new classes, and creation of new annotation with CVAT¹.



Mask

Application of a mask on the GT



Imbalance

$$\frac{1}{\log \left(1.03 + \left(\frac{n^{\circ} class}{n^{\circ} total} \right) \right)}$$

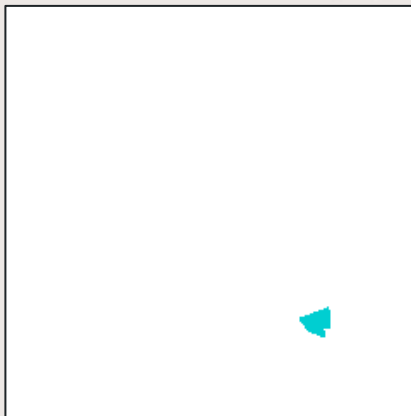
¹ <https://www.cvat.ai>

GT Problems

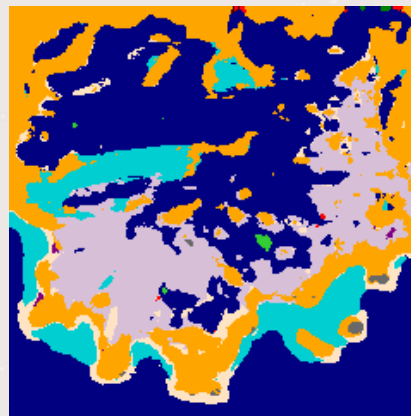
**Poorly
Annotated
GTs**



**Misleading
Results**



**High
Performance**



GT Correction

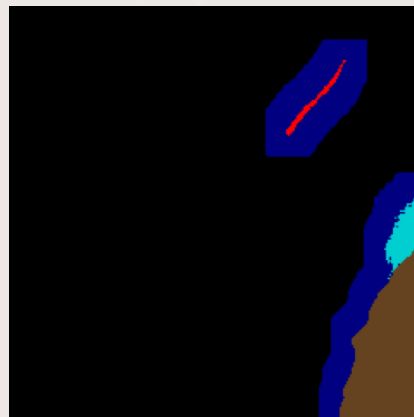
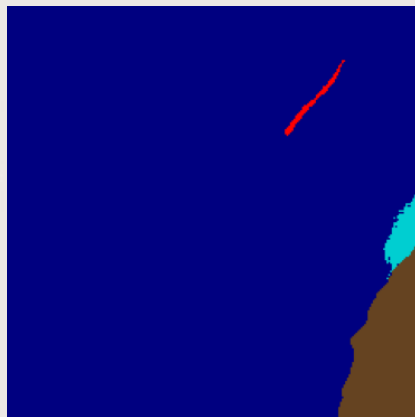
**Fully
Annotated
GTs**

**Correct
Annotation**

**True
Performance**



Mask-Based Cropping



03

Models

MariNeXt & Ticino



MariNeXt (Benchmark)

It is based on **SegNeXt**, an encoder-decoder architecture inspired by **ViT**, utilizing multiscale convolution instead of self-attention, with three variations applied.



**1st-Stage Higher
Resolution
Features**



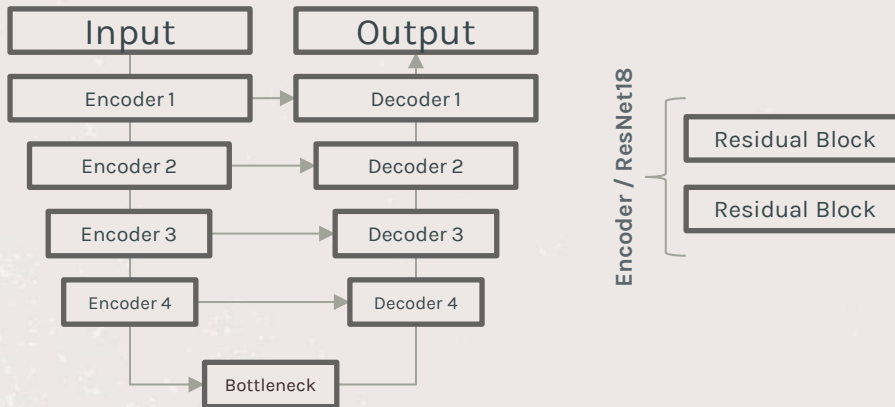
**Very Simple
Copy-Paste
(VSCP)**



**Test-Time
Augmentation
(TTA)**

Ticino

The TICINO model has a U-shaped architecture designed for semantic segmentation. In particular, we use the Segmentation Models PyTorch framework¹, which integrates an encoder-decoder network with ResNet18 as the backbone.



Parameters

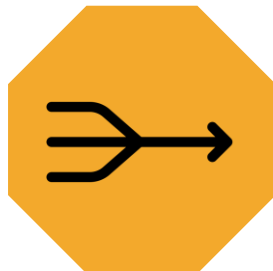
- Epoch = 80
- Batch size = 5
- Learning rate = $2 \times 10^{-4} / 2 \times 10^{-6}$
- Optimizer = Adam
- Loss Function = Binary Cross Entropy

¹https://github.com/qubvel-org/segmentation_models_pytorch

04

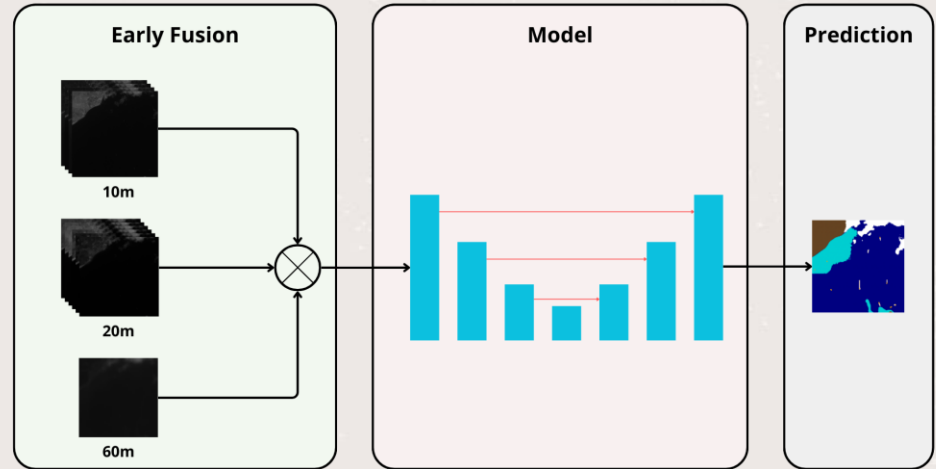
Fusion Strategies

Early, Middle & Late
Fusion



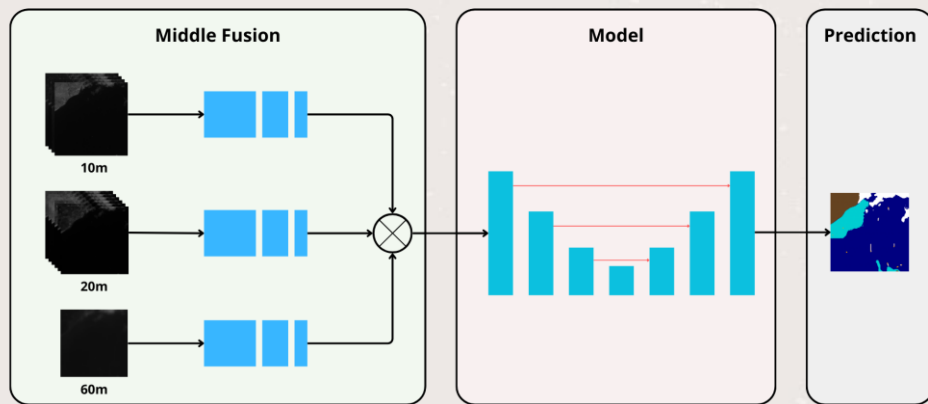
Early Fusion

Concatenating all the modalities into a single representation before being used as input for the model. Specifically, the three modalities are: the bands with a spatial resolution of 10 meters per pixel (blue, green, red, NIR), those at 20 meters per pixel (VNIR, SWIR), and finally those at 60 meters per pixel (aerosols).



Middle Fusion

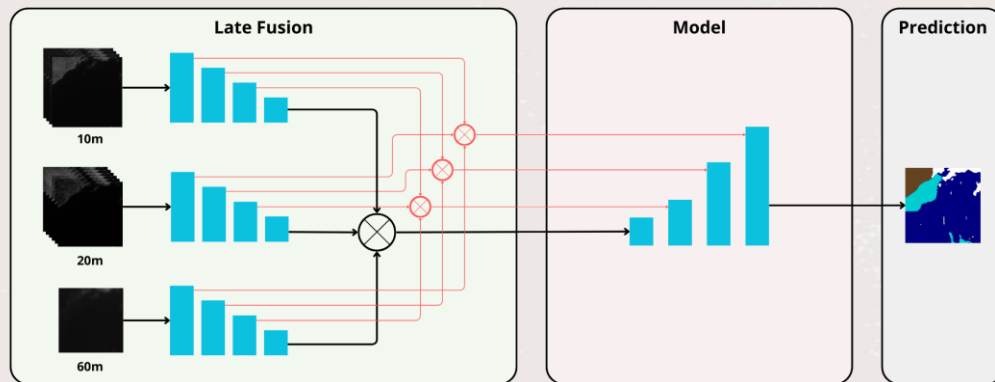
In this approach, the inputs are initially processed separately to extract high-level features from each. Specifically, three convolutional layers are used, each followed by a ReLU activation layer, with respective dimensions of 128, 64, and 32. Subsequently, these features are concatenated to create the input for the model.



Late Fusion

Requires the triplication of the encoder for each of the three modalities. Subsequently, the resulting features are concatenated and passed to the decoder stage. Three variants of late fusion:

- **Late Fusion Standard (LFS)**
- **Late Fusion Variant (LFV)**
- **Late Fusion without Skip Connections (LFnSC)**



05

Results

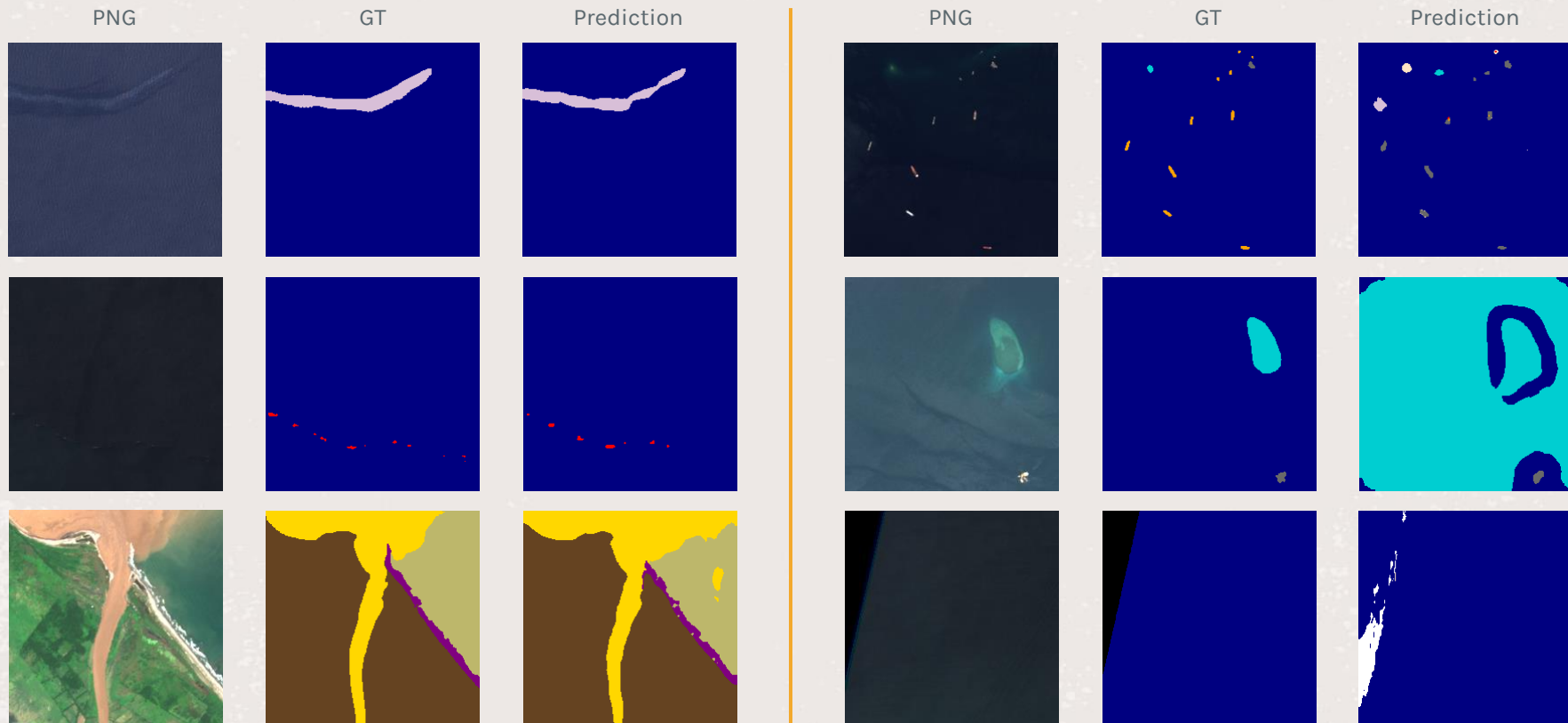
Compare Results and
Predictions



Performance Comparison

	MariNeXt	Early Fusion	Middle Fusion	LFS	LFV	LFnSC
IoU	43.0 $\pm$ 0.4	29.7 $\pm$ 2.3	27.8 $\pm$ 1.3	32.7 $\pm$ 2.7	41.2 $\pm$ 1.2	26.8 $\pm$ 0.9
Precision	60.0 $\pm$ 0.7	40.3 $\pm$ 4.1	36.1 $\pm$ 2.3	42.0 $\pm$ 3.5	56.9 $\pm$ 1.8	35.4 $\pm$ 2.0
F1-score	55.6 $\pm$ 0.6	41.6 $\pm$ 2.4	38.5 $\pm$ 1.6	44.6 $\pm$ 2.9	53.3 $\pm$ 1.3	38.3 $\pm$ 0.8

Prediction with the best model



06

Conclusions

Final reviews and
Future work



Final reviews and Future work



Annotation

New annotations, but to be verified with experts



Fusion Strategies

The idea was successful, let's continue with this strategy



Imbalance

New strategy for class imbalance



Image Augmentation

More images for minor features



Upsampling Techniques

Advanced methods and improving the quality of images



Collaboration

Collaboration with the designers of MADOS and MariNeXt



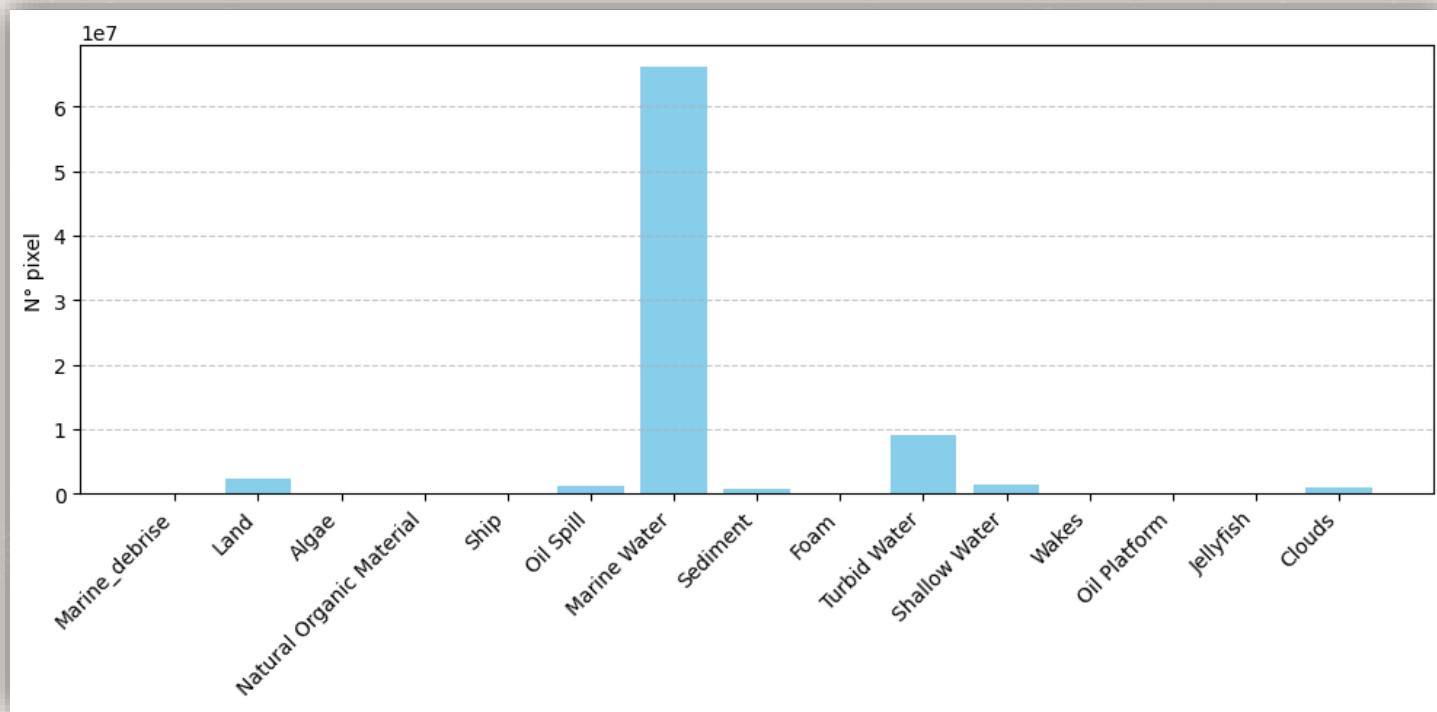
Thank you!

**Do you have any
questions?**

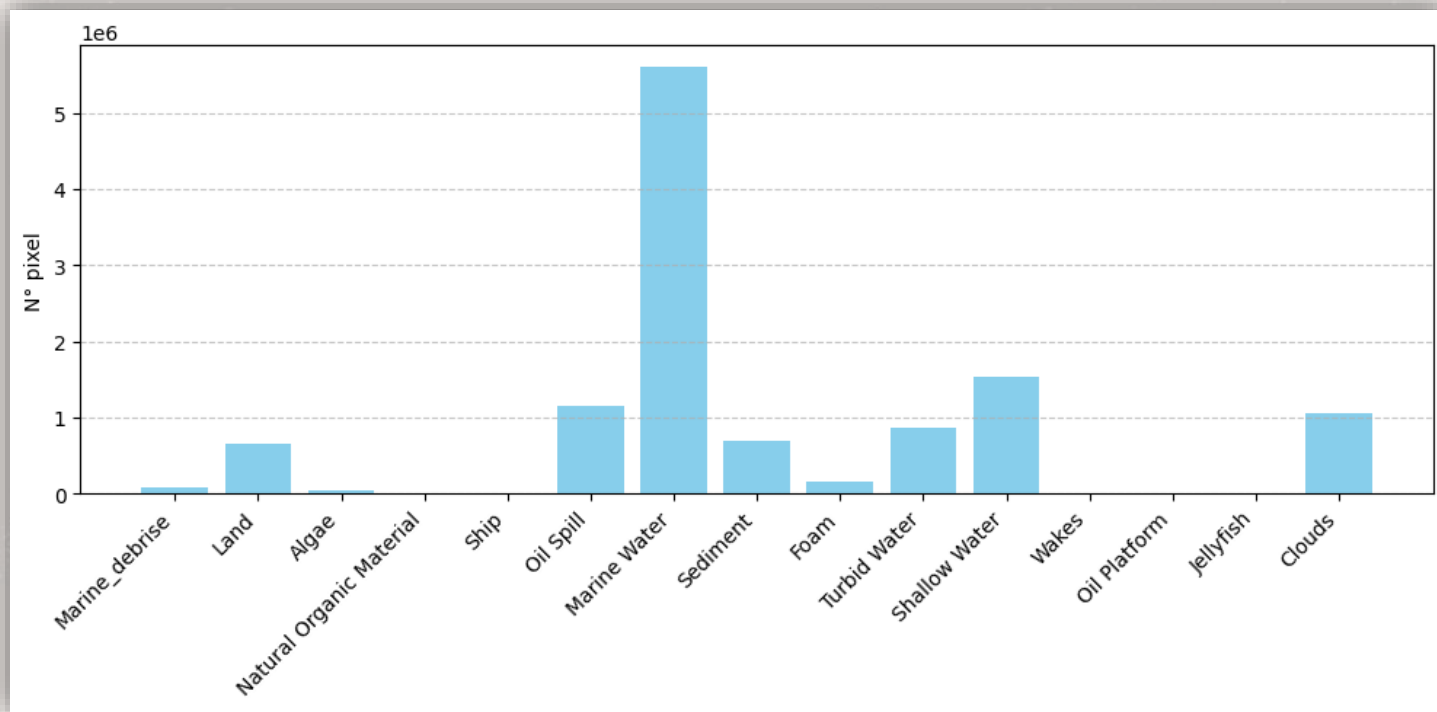
Countries in Consideration



Class Balancing



Class Balancing - Mask



Upsampling technique

Nearest Neighbor

(2x2)

A	B
C	D



(4x4)

A	A	B	B
A	A	B	B
C	C	D	D
C	C	D	D

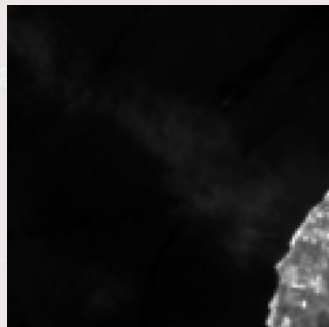
Bilinear Interpolation

10	20
30	40



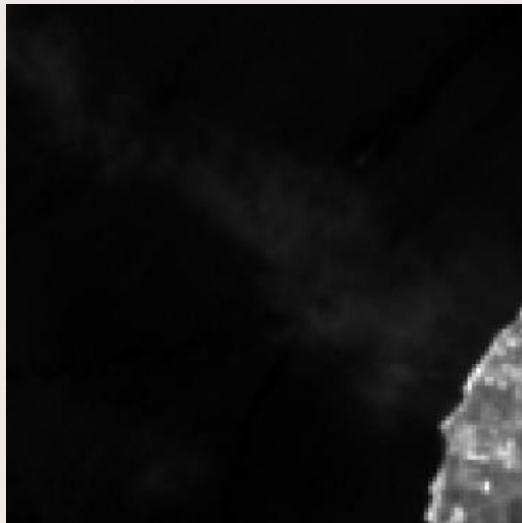
10	12.5	17.5	20
15	17.5	22.5	25
25	27.5	32.5	35
30	32.5	37.5	40

Upsampling technique



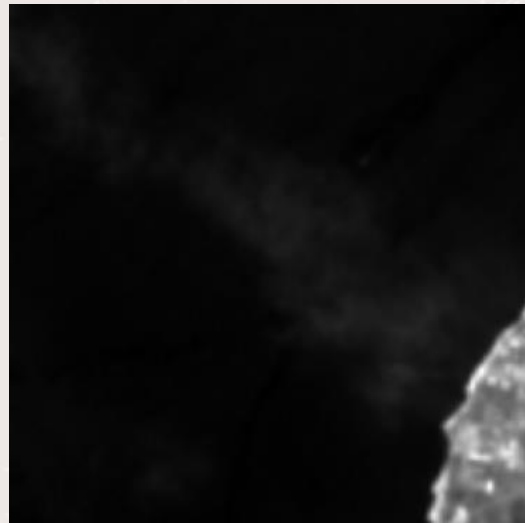
120X120

Nearest Neighbor



240X240

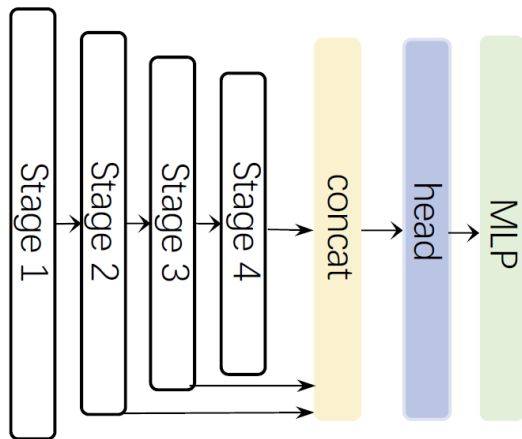
Bilinear Interpolation



240X240

SegNeXt

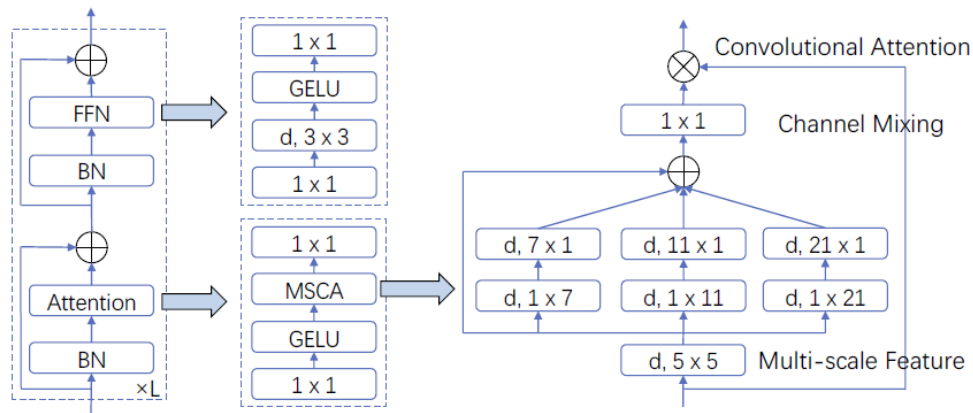
It is an encoder-decoder architecture with a four-stage encoder called **MSCAN**, which uses down-sampling steps and ViT-inspired blocks with multiscale convolution with activation instead of self-attention, while the decoder uses a module called Hamburger to capture global context by decomposition of low-rank matrices.



MSCA

The MSCA is composed of three main components:

- Depth-wise convolution: aggregates local information to keep computational cost low.
- Depth-wise multi-branch strip convolutions: Capture multi-scale context using multiple branches, each with a different-sized kernel.
- Convolution 1×1 : The output of this convolution is used as an attention map to reweight the input of the MSCA.



Hamburger

Lower
Bread

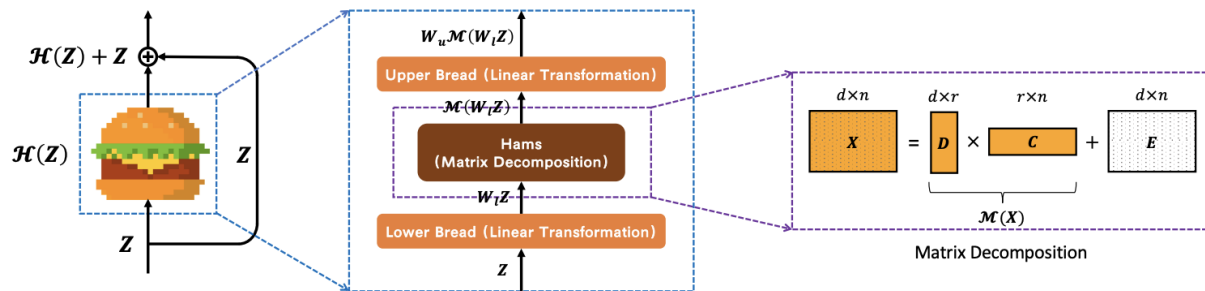
This transformation prepares the data for the decomposition module.

Ham

The matrix decomposition is represented as $M(X) = DC$, where D is the dictionary matrix and C represents the coefficients.

Upper
Bread

Restore the data to its original size



1st-Stage Higher Resolution Features

Modification of SegNeXt that **discards the features of the first stage**, as it produces **prediction maps of size $H/8 \times W/8$** , which are stretched via nearest-neighbor to match the input image. However, in this case, **high-resolution features can be crucial** for effectively modeling floating materials such as marine litter. Taking advantage of these additional features allows the **model to produce prediction maps of $H/4 \times W/4$** dimensions, improving performance in mapping the marine environment under consideration.

Very Simple Copy-Paste (VSCP)

Used for new image construction. Given pairs of input and annotated images, VSCP combines the input and annotated pixels of two images to create a new synthetic input pair that increase the class co-occurrence of the input data.

Very Simple Copy-Paste (VSCP)



Test-Time Augmentation (TTA):

This strategy is to apply several transformations to the input image on the test data and aggregate the results through majority voting, resulting in a more robust and accurate final prediction.

Multimodal Fusion Strategies



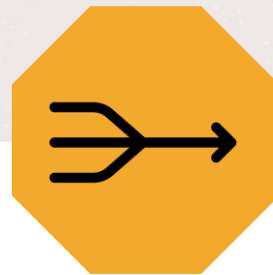
Early Fusion

Concatenating the modalities before being used as input for the model.



Middle Fusion

The modalities are initially processed separately to extract high-level features

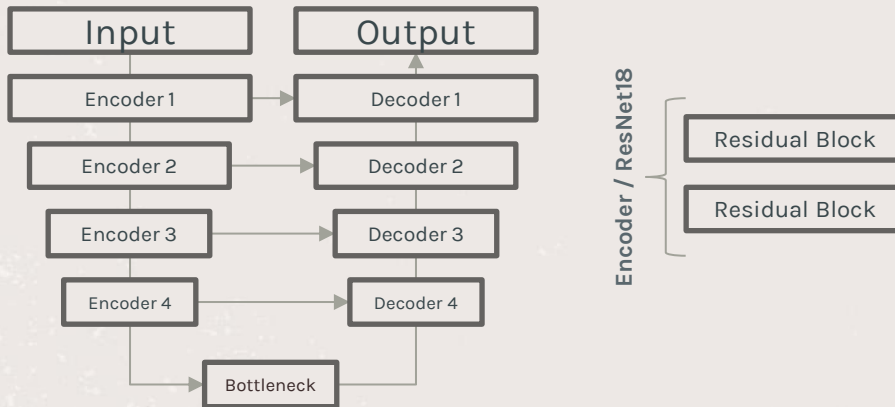


Late Fusion

Triplication of the encoder for each modalities

Ticino

The TICINO model has a U-shaped architecture designed for semantic segmentation. In particular, we use the Segmentation Models PyTorch framework¹, which integrates an encoder-decoder network with ResNet18 as the backbone.



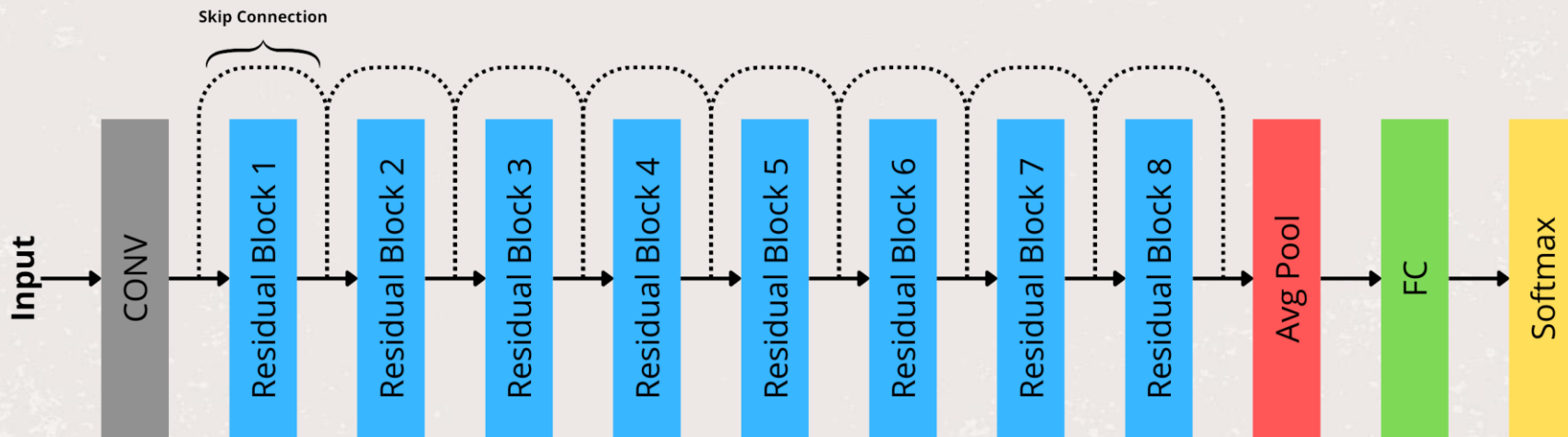
Parameters

- Epoch = 80
- Batch size = 5
- Learning rate = $2 \times 10^{-4} / 2 \times 10^{-6}$
- Optimizer = Adam
- Loss Function = Binary cross entropy loss

¹https://github.com/qubvel/segmentation_models.pytorch

ResNet18

The TICINO model has a U-shaped architecture designed for semantic segmentation. In particular, we use the Segmentation Models PyTorch framework¹, which integrates an encoder-decoder network with ResNet18 as the backbone.



Performance comparison

Marine Debrise	MarineXt	Early Fusion	Middle Fusion	LFS	LFV	LFnSC
IoU	26.2 $\pm$ 0.6	16.5 $\pm$ 3.7	21.4 $\pm$ 1.2	17.7 $\pm$ 5.1	25.4 $\pm$ 2.2	12.0 $\pm$ 4.3
Precision	30.6 $\pm$ 1.7	17.1 $\pm$ 3.7	22.3 $\pm$ 1.5	18.2 $\pm$ 5.3	26.7 $\pm$ 2.4	12.3 $\pm$ 4.5
F1-score	41.5 $\pm$ 1.4	28.2 $\pm$ 5.5	35.2 $\pm$ 1.6	29.7 $\pm$ 7.5	40.4 $\pm$ 2.8	21.2 $\pm$ 6.9

Performance comparison

Oil Spill	MarineXt	Early Fusion	Middle Fusion	LFS	LFV	LFnSC
IoU	15.7 $\pm$ 3.8	4.5 $\pm$ 1.5	2.6 $\pm$ 0.7	3.6 $\pm$ 0.7	7.6 $\pm$ 1.9	3.8 $\pm$ 1.1
Precision	16.4 $\pm$ 4.2	4.6 $\pm$ 1.5	2.6 $\pm$ 0.7	3.6 $\pm$ 0.7	7.7 $\pm$ 1.9	3.8 $\pm$ 1.1
F1-score	26.9 $\pm$ 6.0	8.6 $\pm$ 2.7	5.0 $\pm$ 1.4	6.9 $\pm$ 1.3	14.0 $\pm$ 3.3	7.2 $\pm$ 2.1